

Class 1.6 — Control Box II

Sunday, July 19, 2026 · 3 hours · Mr. Holcombe's house

THE ONE GOAL

The control box exits 1.6 fully soldered, with the power wire built, the tether terminated, the fuse re-sized from measured FLA, and every pad meter-checked clean. Signed off in the engineering notebook. Ready for the smoke test at 1.7's pre-pool review.

BEFORE YOU TEACH THIS

- Watch the MATE "Creating your ROV's power wire" and "Connecting the Tether to the Control Box" presentations end-to-end the night before.
- Read pages 319–323 (§7.5 — Power) and pages 466–473 (§9.4 — Tether) of the Underwater Robotics textbook. The fuse-sizing math lives in §8.7.4 (pages 425–428).
- Verify the 12V SLA battery is fully charged and the charger is operational. The battery is the gating tool for tonight's fuse-selection measurement.
- Print ELEC-NRD-003 (MATE's fuse-sizing safety rule). The engineering-notebook entry tonight cites it by number.
- Practice the power-wire build yourself the night before. Stripping, crimping, heatshrinking — if your fingers are out of practice, the kids will see it.
- Internalize Section 4.3 again. The fuse-selection math is THE judge-magnet at safety inspection in April. The kids must own that measurement, not the advisor.

WHY THIS CLASS MATTERS

1.5 was the bridge from practice to the real thing. 1.6 closes the box and stages the smoke test. The discipline tonight — meter-checking every pad, sizing the fuse from measurement, signing the board off in writing — is what protects every dive between now and the regional in April.

There is a specific competition-rule angle here: MATE safety inspection asks teams to cite their fuse selection against the measured full-load amps. Teams that have that measurement documented in their engineering notebook pass clean. Teams that guessed the fuse size lose points and sometimes can't dive. We start that habit tonight, on a board the kids built themselves.

HOW THE MEETING RUNS — BLOCK BY BLOCK

0:00 – 0:15 · Welcome and the safety brief

Start on time. Read the agenda end-to-end, hand the standing safety brief to the assigned Safety Briefer, then run the four questions and the today's-brief card. Slow down on the new safety beat: BATTERY STAYS DISCONNECTED until the pre-power-up review at the end of class. The kids should hear that line twice, once in the brief and once at the start of the power-wire block.

Assign the four rotating roles right here: Notetaker, Safety Briefer, Photographer, Timekeeper. The Photographer matters tonight — the pre-power-up meter-check is photo gold for the engineering notebook and the eventual marketing display.

0:15 – 0:30 · Where the board stands

Inventory what 1.5 closed and what's open. Read the next block of components from the SeaMATE Guide as a team — one reader, others listen. No iron picked up yet. This is the orientation block.

Name the discipline that carries from 1.5: working vs. crafted. Every joint tonight gets meter-checked at the bench before we move on. If you find a 1.5 joint that fails inspection, re-flow it now, not later.

0:30 – 1:30 · Finish the control-box wiring

Sixty minutes hands on. The build loop is unchanged from 1.5: read the slide together, stage the components, solder one student at a time, inspect, meter-check, next slide. Hold that rhythm.

Your job is to observe and photograph. Do not pick up an iron. If a kid is stuck, ask one diagnostic question and step away. The judges in April will ask each student to explain how they did their joints; the team that hears "I did this one" gets the points.

1:30 – 1:40 · Break

Ten minutes. Iron tips in holders, kids step away. Resume at 1:40 sharp.

1:40 – 2:25 · ROV power wire + tether termination

Two parallel tasks. One team builds the power wire — strip, crimp ring terminals at the battery end, heatshrink the conductors, run them through strain relief at the control-box end. Another team terminates the pre-terminated tether at the control box per the Guide. Rotate at minute 22 so every kid touches both tasks.

Polarity verification at the power-wire termination is the moment most likely to be done wrong by tired hands. Two people verify — one reads the silkscreen at the board, the other reads the wire — before any commit. Reversed polarity here ends the smoke test before it starts.

2:25 – 2:40 · Fuse selection from measured full-load amps

This is the section that wins safety inspection in April. Battery connected to the watt meter only (NOT the board). Measure full-load amps — at this stage, the highest current draw the team can simulate with what's on the bench, plus a documented margin. Round up to the next standard blade-fuse size: 3 / 5 / 7.5 / 10 / 15 / 20 A.

The 3 A practice fuse comes out; the measured-and-rounded fuse goes in. The math, the chosen size, and the rule citation (ELEC-NRD-003) all go in the engineering notebook tonight. Photograph the page next to the fuse holder. That photo is judge bait in the best way.

2:40 – 3:00 · Pre-power-up meter-check (three reads × every pad)

Bring the team together at the bench. Walk every pad with the multimeter, in order, against the schematic. Three reads on each: continuity to next pad, no shorts to ground or neighbors, polarity confirmed where applicable. Silence at the wrong place means re-flow or re-route — tonight, not next week.

When every pad passes, the kid who ran the final meter-check signs the engineering notebook page. Date, time, results. That signature is the answer when a judge asks "who verified this?" Then the wrap-up slide and the tool checkout. We do NOT power up tonight. The board exits 1.6 ready for the smoke test at the top of 1.7.

THE PARENT LANE — SECTION 4.3

The fuse-selection math is THE judge-magnet at safety inspection in April. The kids must own that measurement — write down the FLA reading, do the round-up, cite the rule, sign the notebook. You photograph; you do not write. If you find yourself reaching for the pen, you've already lost the section. Step back. Ask one diagnostic question, then wait.

COMMON PROBLEMS & QUICK FIXES

Symptom	Quick fix
Power wire won't accept the ring terminal cleanly	Strip back another 1/8 inch, re-crimp. If still loose, the wire gauge and terminal don't match — substitute the correct terminal.
Heatshrink won't shrink (or shrinks unevenly)	Heat source too cold or too far away. Hot air gun ~6 inches from the joint, move slowly along the length. A lighter works in a pinch but
Watt meter reads zero with the battery connected	Battery probably died or fuse blown internally. Substitute a known-good battery to isolate.
Multimeter beeps continuously on every probe touch	Probes are shorted to each other inside the meter or the meter is set to a too-sensitive range. Set to 200 Ω , probe the probes — should beep when touching.
Reversed polarity discovered at the power-wire termination	Stop! Do not connect the battery. Undo the heatshrink (cut it off with the flush cutters), re-orient the conductors, re-heatshrink. Two-person
Pre-power-up meter-check finds a short to ground	Verify the shorts visually first — most are solder bridges between adjacent pads. Add flux, re-heat both pads, wick with desoldering braid. Re-meter.
WHAT "DONE" LOOKS LIKE	Do not power up until the short clears.

- Control box fully populated per the SeaMATE Guide. Every joint shiny, smooth, concave.
- Power wire built with ring terminals at the battery end, polarity-verified, strain-relieved at the control-box end.
- Pre-terminated tether landed at the control box; every conductor continuity-checked end to end.
- Fuse re-sized from measured full-load amps; the math + the chosen size + ELEC-NRD-003 citation in the engineering notebook.
- Pre-power-up meter-check passed every pad. Signed off in the engineering notebook by the student who ran it.
- Action-items slide filled in and read back before the closer. Pool prerequisites for 1.7 named with owners.

IF YOU FALL BEHIND

Two protect-the-priority rules. (1) Never skip the pre-power-up meter-check. If the wiring isn't finished, leave the unfinished part for a brief mid-week catch-up and meter-check what IS done before sign-off. Powering an unverified board next week is the only worse option. (2) Never skip the fuse-selection notebook entry. Even if the math is a rough first pass, getting it ON THE PAGE is what protects April. We refine in 2.x.